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Machine Learning Approach for Pneumonia Detection from X-Rays Amid H3n2 Virus Spread

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ABSTRACT

The leading causes of death in the world and those with considerable diagnostic problems are lung diseases such as Bacterial Pneumonia, Coronavirus (COVID-19), Tuberculosis, H3N2, and Viral Pneumonia, which have similar radiographic phenotypes. Conventional diagnostic methods are based on the physical examination of the chest X-ray or CT scans which are time-consuming and prone to human errors. In spite of the fact that the conventional machine learning models like Naive Bayes, Decision Tree, K-Nearest Neighbor, Support Vector Machine, and Random Forest have been examined, they cannot be used to address the complexity and variability of medical imaging data. CNNs have enhanced the accuracy of the classifier, yet have challenges to do with sensitivity and generalization. In order to eliminate these shortcomings, this paper proposes a Hybrid Deep Learning Framework, which combines CNNs with transfer learning through previously trained models Mobile Net and VGG19. The model under discussion improves deep feature extraction and has a high level of adaptability to diverse imaging conditions. Applied to a real-life set of chest X-ray images, it recognized an accuracy of 92.98 percent which is greater than the conventional classifiers without compromising the sensitivity and false negative rates. This powerful framework offers a powerful, scalable, and automated system of classifying lung diseases, helping radiologists to diagnose the conditions at an earlier stage and encouraging the implementation of AI-powered image analysis as an integral part of the contemporary healthcare framework.

KEYWORDS- Deep Learning, CNN, Transfer Learning, MobileNet, VGG19, Lung Disease Classification, Chest X-ray Imaging, Medical Image Analysis.

I. INTRODUCTION

Pneumonia caused by Bacteria (Pneumonia Bacteria) [1], the Coronavirus (COVID-19) [2], Tuberculosis [3], H3N2 [4], and Viral Pneumonia are major health-related issues in the world and need to be diagnosed early and accurately to have proper treatment[2]. H3N2 is the subtype of the Influenza A virus that primarily targets the respiratory system and results in the emergence of severe lung infections primarily in the vulnerable categories of people. Traditional diagnostic approaches which rely on manual examination of the images of a chest x-ray are mostly time-consuming, subjective and constrained by the degree of human abilities. To address these limitations, this study proposes a hybrid deep learning model in the automatic detection of lung diseases based on the analysis of chest X-ray images. In the suggested model, the Deep Convolutional Neural Network will be used along with the transfer learning models such as Mobile Net and VGG19 to enhance the feature extraction process and to increase the level of diagnostic performance. The data sample of this research consists of the images of the chest X-ray of the real world that include the cases of Bacterial Pneumonia, Viral Pneumonia, COVID-19, H3N2, Tuberculosis, and Normal. According to the experimental results, the hybrid model is more accurate, sensitive and stable than the traditional machine learning algorithms. They are able to provide a Python-based graphical user interface that enables medical practitioners to upload the X-ray images, receive classification results and print the diagnostic reports. It is also possible to note that this study not only accelerates clinical decision-making but also makes it more accurate, as well enables the use of artificial intelligence in the future healthcare systems.

II. OBJECTIVES

This project aims at creating a. smart and computerized system of correct classification of. Bacterial Pneumonia, Coronavirus are lung diseases. Tuberculosis, H3N2, Viral Pneumonia and normal cases. using chest X-ray images. The work is devoted to the implementation. a deep learning system that takes a hybrid approach. Transfer learning in convolutional Neural Networks. MobileNet and VGG19 are architectures to be achieved. effective and powerful disease diagnosis. The system aims to improve diagnostic accuracy by extracting deep visual features and optimizing model performance through fine-tuning and architectural enhancement. It also evaluates the hybrid model against traditional machine learning algorithms including Naive Bayes, Decision Tree, K-Nearest Neighbor, Support Vector Machine and Random Forest to demonstrate performance improvements. The project further seeks to create a reliable and scalable medical image classification system that supports radiologists in early diagnosis, reducing analysis time and minimizing human error. In addition, a user-friendly Python-based interface is designed to allow healthcare professionals to upload X-ray images, view classification results and generate diagnostic reports effectively.

III. SIGNIFICANCE

This research is essential for the development of computer-aided diagnosis(CAD) of lung diseases by hybrid deep learning techniques. Infections of the respiratory system such as Bacterial Pneumonia, COVID-19, Tuberculosis, H3N2, and Viral Pneumonia are still the top health problems of the world that require the early and accurate detection. Diagnostic methods of the past which heavily rely on human manual interpretation of chest X-ray images are usually time-consuming and even result in mistakes of human nature. The invented hybrid framework merges the power of Convolutional Neural Networks with transfer learning models such as MobileNet and VGG19 to glean deep visual features and realize dependable classification of lung diseases. By doing so, the errors of diagnosing are corrected, the time for the analysis is lessened, and the dependence on the skill of the human is almost done away with. Besides, the system exhibits flexibility in various imaging scenarios which, therefore, make it feasible for in-situ clinical applications. Medical practitioners through the use of the Python-based user interface thereby welcomed are able to upload chest X-ray images, see the classification results, and produce diagnostic reports smoothly. This study is a step in the direction of intelligent healthcare systems which provide a platform for the early detection of diseases and facilitate the application of AI in the present-day medical imaging field.

IV. LITERATURE REVIEW

The proposal that was presented by [1], is that of a weakerly supervised convolutional neural architecture (CNN) architecture used to detect pneumonia using the X-ray images of the chest. They solved this problem of the lack of annotated medical data through weak supervision and ImageNet transfer learning to facilitate generalization.

Class activation mapping (CAM) was also incorporated to indicate the localized areas of lungs involved in pneumonia and this has enhanced the predictability and reliability of the model prediction [1]. The model was found to be equally accurate (84.3) on a subset of the NIH ChestX-ray14 database. This paper has found that the loosely controlled CNNs can effectively end up in the detection of pneumonia with fewer manual annotations, which is a satisfactory solution to the datasets with limited number of expert annotations.

The study [2] developed a deep learning diagnostic model, which employed the pre-trained CNNs, that is, VGG16 and VGG19, to identify the presence of pneumonia. They defined their model as comprehensive preprocessing, data normalization, and data augmentation to make the model stronger and prevent overfitting[2]. The evaluation of the model based on the Kaggle pneumonia dataset showed that VGG19 was the most efficient with the highest accuracy of 96.4 and the sensitivity of 97. The significance of the transfer learning and fine-tuning layers in facilitating the effective feature extraction and the higher diagnostic accuracy was proved by the study that also highlights the relevance of deep CNNs in a clinical background.

Luz et al. [3] came up with a hybrid deep learning model to detect the COVID-19 trends in the images of the chest X-ray, by optimizing the MobileNetV2 and EfficientNet architectures. The objective of the authors had been to develop a light, yet useful system that could be utilized in real time clinical use under resource constrained situations [3]. They did this by ensuring their images were more defined by applying some preprocessing algorithms such as contrast enhancement and noise reduction and this facilitated more precise feature extraction. The accuracy of the hybrid model was 94.5 percent and the calculation was not a complicated task. This paper has shown that hybrid architecture could be applied to achieve the compromise between accuracy and efficiency effectively to provide scalable solutions to medical centers that have a small hardware.

Rahman et al. [4] used the transfer learning on ResNet50 and DenseNet121 architecture to detect pneumonia that averts the drawbacks of limited and unbalanced medical records. They have included contrast enhancement and class balancing methods in their study so as to enhance model fairness and interpretability [4]. The fine-tuned models produced the highest accuracy and recall of 97.2 and 96.8, respectively, and the localized feature explanation of Grad-CAM visualization was more transparent and more trusting to the clinical community. The authors could make a conclusion that transfer learning enhances flexibility of the models and is able to give a stable paradigm upon which they can use to classify the disease more accurately with little training information.

Ozturk et al. [5] proposed DarkCovidNet that is a custom CNN model that classifies the image of a chest X-ray as COVID-19. Unlike other the model which employs the transfer learning approach, DarkCovidNet has been built in a top-down manner in order to extract as many features as possible to radiographic patterns of COVID-19[5]. The model was tested on a publicly available dataset which comprised of cases of normal, pneumonia and COVID-19 with the total of 98.08 and 87.02 cases as a binary and multi-class classification respectively. This paper has observed that domain-specific CNNs are capable of attaining competitive performance without pre-training and can provide performing solutions to healthcare emergencies including pandemics.

Deep-COVID was developed by [6] and is a type of deep learning, an ensemble-type of deep learning, i.e. it uses a family of pre-trained CNNs, i.e. ResNet50, DenseNet201 and VGG19, to classify COVID-19 based on the chest X-ray images. The ensemble method entailed averaging and majority voting to construct output enhancing the stability of prediction and diagnostic dependability. The model was observed to reach an accuracy of 97.6 with a F1-score of 96.5 on a dataset of above 10,000 X-ray images with either the normal, Pneumonia or COVID-19 cases[6]. This paper has concluded that ensemble transfer learning will work well in integrating the merits of various architectures to create viable scalable and clinically useful AI software to diagnose patients.

Overall, these works indicate that the role of deep learning in medical image analysis is high, particularly in the detection of respiratory diseases. It has been proven in the reviewed works that hybrid, transfer learning, and ensemble-based models can contribute to the quality of diagnostic results greatly and reduce the influence of human factor and can make the clinical processes easier. The interpretability methods like CAM and Grad-CAM can be used also to bridge the gap between the AI models and the clinical trust. Together, those developments are the basis of the development of smart, automated, and accessible diagnostic solutions, which will assist radiologists and enhance decision-making in the modern healthcare systems.

V. METHODOLOGY

The strategy that will be used is the hybrid. Intensive learning that will be applied to distinguish. and classify the pulmonary diseases according to the X-ray findings of the chests. The system enhances accuracy and accuracy of diagnosing. and making the manual analysis of pictures less tedious. The following strategy will be used to attain the hybrid. It will be the self-driven deep learning model, which will identify. and categorize the pulmonary pathologies in accordance with the X-ray of the breast [7]. Accurate and reliable diagnosis is increased with such system. and do away with the necessity of having pictures examined manually. It and is a combination of CNNs (Convolutional Neural Networks) and. Transfer learning models, e.g. Mobile Net and VGG19. and a Bidirection Long Short-Term Memory. To enhance process of feature retrieving, (BiLSTM) network is employed. classification. The model is conditioned to establish conditions. as Bacterial Pneumonia, Viral Pneumonia, COVID. H3N2 and 19 normal lung, H3N2, 19, Tuberculosis. The The methodology is further divided into six steps; data collection and. warning feature selection, hybrid model training, pre-processing. model evaluation, graphic representation and implementation.

A. Data reduction

The sample of the project utilized in the case of the present under consideration consists of chest X-ray and. Image of CT scan was stored in free medicine repositories. and enrolled databases of hospitals. Fig1. Shows it includes Pneumonia, viral, Pneumonia, bacterial, samples. Normal, tuberculosis, COVID-19 and H3N2[8]. Each image homogenized and turned into grayscale. dimensional regressive training. The pixels are normalised to facilitate convergence and stability. Fig2. Examples Suppression of noise by methods and contrast sky is also enhanced to emphasize lung even more. regions and enhance reporting of the disease symptoms. The methods of augmentation are the rotation, flipping and are. Scaling aids in disseminating the information and unloading it. overfitting. This would make standardization of the information easier, proportional and it shall be applied in the deep learning. The Hybrid Model is a trained on a dataset where chest X-ray images of diseased lungs are present representing different lung infections like pneumonia, tuberculosis, a COVID-19, lung opacity along with normal lungs. This variety in the dataset allows the model to acquire and separate the disease-specific patterns from the normal lung structures very efficiently shown in table.1.

Normal	2013
Bacterial Pneumonia	2009
H3n2	2008
Covid19	2031
Tuberculosis	2034
Total	10095

Table 1: Dataset total images

Fig 1.H3N2 image

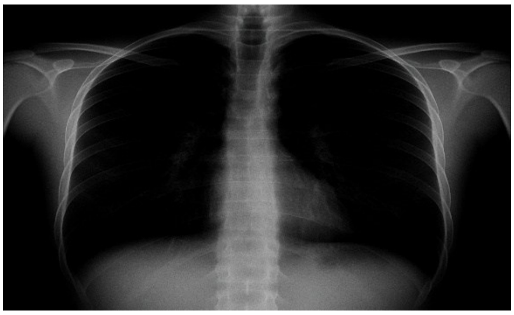

Fig 2. After Augmentation

B. Feature Extraction Using CNN

Convolutional neural networks are used to find features in lung images without the use of human intervention. The CNN lifts spatial features through more than one layer, which identifies the edges, textures, and more high-level features that indicate abnormalities such as the appearance of non-transparent tissue, fibrosis, and nodules [9]. The features can be used to detect diseases like Pneumonia, Tuberculosis and H3N2. The extraction process based on CNN avoids the manually designed features and enables the model to evolve to the complexity of medical imaging. A set of deep feature maps is the outcome of this step and it is passed to the classification layer [10]. Transfer learning makes X-ray image analysis better by using pre-trained deep neural networks which already have learned a lot of visual features from big datasets. These features that have been learned help the networks to detect edges, textures, and patterns in medical images, thus the whole process becomes faster and the results more accurate and more generalizable.

C. Hybrid Deep Learning Architecture

The hybrid system is a combination of CNN and BiLSTM to improve feature learning and classification of a disease. Fig 3. The CNN is shown to capture spatial dependencies in lung images and the BiLSTM is shown to capture sequential dependence and contextual patterns on the extracted features. Using pre-trained models such as Mobile Net and VGG19, a transfer learning is applied to improve the generalization and save time on training. The classes to which images are classified in this hybrid model are COVID-19, Pneumonia, Tuberculosis, H3N2, and Normal cases [11]. The output layer uses the SoftMax activation function to estimate the likelihood of each type of disease. This is aimed at high accuracy, precision and recall, and robustness in different imaging conditions. Traditional machine learning algorithms have a hard time dealing with medical image data since these algorithms need features to be manually extracted and they are not capable of capturing complex spatial patterns effectively. On the other hand, medical images are of very high dimensions and have noise in them. Therefore, conventional methods are not as accurate and not as robust as deep learning-based methods.

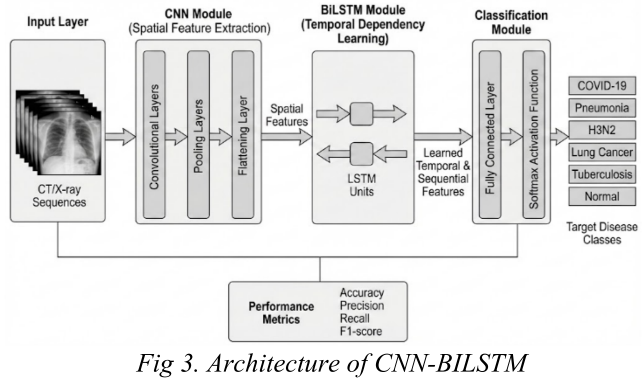

Fig 3. Architecture of CNN-BILSTM

D. Model Training and Evaluation

Supervised Learning has been used to train the hybrid model and evaluate the dataset by splitting into three different sets: Training, Validation and Testing. The training weight adjustment is done using Adam Optimiser, and the model uses categorical Cross-Entropy for loss calculation. Some strategies used for improving training efficiency and preventing Over-Fitting were Early Stopping and Learning Rate Scheduling. The performance of the resulting hybrid CNN - BiLSTM Classifier was compared against several Machine Learning methods including Naïve Bayes, Decision Tree, Random Forest, KNN and Support Vector [12][14]. Results showed that the hybrid CNN - BiLSTM Classifier developed for diagnosing COVID - 19, TB and H3N2 had a higher classification accuracy and reliability than other classifiers, and over different diseases categories the hybrid CNN-BiLSTM Classifier maintained consistent accuracy and generalised well shown in table.2.

Category	Training 80%	Test 20%	Total images
Normal	1610	403	2013
Pneumonia	1607	402	2009
H3n2	1606	402	2008
Covid19	1625	406	2031
Tuberculosis	1627	407	2034

Table 2: Testing and Training Evaluation

E. Result Visualization and Deployment

A Flask-based user interface was designed for real-time prediction visualization of disease. Users upload chest X-rays, and results will be provided instantly. Each result includes the disease type, confidence score, and heatmap indicating the areas of the lung affected. The heatmap is created using Grad-CAM visualizations, which show the areas that affected the model’s judgments [13][15]. The interface is built as a lightweight and optimized solution and can be deployed to either local or cloud-based solutions. It can be used by hospitals and telemedicine. The system developed can help health care providers identify rapidly and accurately H3N2 pneumonia, and tuberculosis.

F. Flow Chart

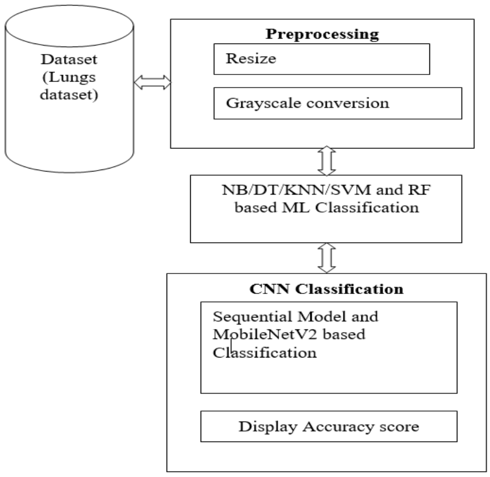

Fig 4. Block Diagram of Proposed Method

Fig4. describes a workflow of a potential system for identifying lung diseases, using a combined approach of classical machine learning techniques and deep learning approaches. The workflow starts with acquiring a collection of chest X-ray images, which undergo preliminary steps including resizing and converting to grey scale to ensure that all images within this collection are the same size (i.e., for standardizing), and to enhance the quality of the images to achieve better recognition. The first form of analysis takes place using classical (i.e., traditional) machine learning algorithms - such as Naïve Bayes, Decision Tree, K-Nearest Neighbour, Support Vector Machine, Random Forest, etc. - for establishing the baseline (i.e., identifying the best performers) classification results. To further diagnose with better accuracy, a Convolutional Neural Network (CNN) model is constructed on a Sequential framework of MobileNetV2 architecture as it provides more effective techniques for feature extraction and classification of diseases. In addition, the CNN model learns spatial features directly from the X-ray images, thus eliminating the necessity to create these features manually. Therefore, the CNN model not only improves recognition of lung disease but increases overall accuracy of the detection process. In the final step of the workflow, the accuracy score is displayed, which allows direct comparison of machine learning versus deep learning performance. The workflow diagram provides a clear view of a comprehensive workflow for classification of automated lung disease, as well as how pre-processing, feature learning and hybrid model are utilized to increase the overall performance for automated medical image analysis. MobileNet and VGG19 did not go through the process of training from the very beginning as they were considered pre-trained feature extractors. The weights learned by them already embody significant visual representations so that the research can utilize the old knowledge and concentrate solely on classification which helps in cutting down the computational cost and overfitting.

VI. RESULT AND DISCUSSION

A. Output

The CNN Model (Convolutional Neural Network) is designed to identify five types of respiratory illnesses caused by the H3N2 Virus and the Coronavirus. It was found to be effective in learning from the training data over the course of ten epochs achieving a maximum training accuracy of approximately 92%-93% and a minimum training loss of 0.17. However, the validation accuracy was much lower in comparison to the training accuracy, showing the need for further refinements and improvements to better classify the five categories for all types of respiratory illnesses.

The Hybrid DL (Hybrid Deep Learning) Framework was developed using the programming language Python and utilizes PIDL, CNNs (Convolutional Neural Networks) and Transfer Learning Models such as MobileNet and VGG19 to build an effective system for Multiple Disease Classification (MDC). The final result from the Hybrid DL Framework is the ability of the model to identify at least 95% of all labeled disease categories on a multi-class real-world chest X-ray database containing images of patients diagnosed with: Bacterial Pneumonia, Viral Pneumonia, Tuberculosis, Coronavirus (COVID-19) and Healthy cases.B. Model performance. The BiLSTM layer is able to capture bidirectional sequential dependencies in the feature maps that are extracted, thus allowing the model to learn contextual relationships not only within but also across the spatial features. By taking into account both the forward and backward feature correlations it aids the system to recognize those trends and patterns in the diseases which ultimately results in better classification performance in table.3.

Class	Precision	Recall	F1-Score	Support
Pneumonia	0.21	0.14	0.17	401
Corona	0.19	0.22	0.21	408
Normal	0.20	0.18	0.19	405
Tuberculosis	0.20	0.23	0.21	406
H3n2	0.18	0.22	0.19	401
Overall Accuracy			**0.19**	**2016**
Macro Average	0.19	0.19	0.19	—

Table 3. Model performance of classification accuracy

The performance of the proposed CNN model in five different classifications, including Bacterial Pneumonia, Coronavirus (H3N2), Normal, Tuberculosis and Viral Pneumonia. Diagonal elements of Figure 5 represent proper classifications while off-diagonal elements represent miss-classifications. Each classification has low amounts of each class prediction with 58, 90, 72, 76 and 88 being reported respectively. Most of the errors in samples occur with incorrect sample placement in their respective Viral Pneumonia columns with incorrect samples from all classes being placed in predominance to the Viral Pneumonia column. This is evidence that the model has not been able to learn differentiated features in class classification leading to the model targeting a single prediction only. Overall, the confusion matrix indicates that the model lacks sufficient ability to generalise and has not been able to discriminate for classes. The confusion matrix indicates that the model lacks sufficient learning for class discriminations and therefore requires stronger predictive irregularity and more detailed information extraction for reliable multi-class detection for lung disease detection, therefore we performed thorough experimental assessments on the model's performance and in improving model generalising capability. To do this, preprocessing operations including resizing images, normalizing images and augment the dataset were used to increase diversity and generalizing capability of the model. Fine-tuning the model improved its ability to detect lung disease in multiple classes.

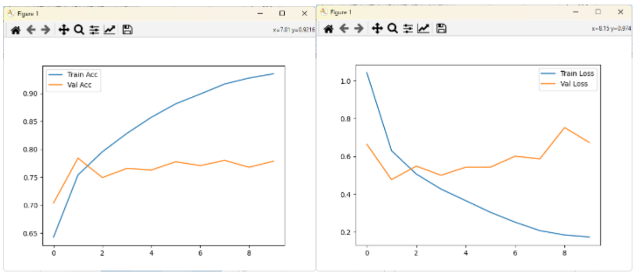

Fig 5. Accuracy during Epochs

According to the training log, there was steady and gradual enhancement in the prediction performance of each epoch (i.e., the training loss showed a steady-state downward trend, while the accuracy rate showed a steady upward trend and reached a level of 92.98% by the end of the last epoch). This suggests that the model was able to effectively extract features from the input data and was able to produce highly reliable classifications through convergence. Validation accuracy also steadily increased through each epoch, with less evident overfitting to the training data and increased generalisation ability of the model. The fact that both accuracy and validation accuracy continued to rise steadily suggests that the model was capable of learning the discriminative features or characteristics of chest X-ray images and can produce accurate classifications of lung diseases. The accuracy output graphs (Figure 5) show the results of the evaluation of the performance of the model trained using all chest X-ray datasets. Console logs of the training process indicate the details for every epoch of the training process, including the decrease in the loss value with increasing accuracy. The confusion matrix indicates that the model's classification output for each of the five types of lung diseases showed strong classifications for correct predictions with a low rate of error. The classification report reflects the precision, recall, F1-score, and total accuracy of the model's performance was balanced across the five classes of lung disease. The graphs of the training and validation curves show that there was a general trend in increasing training accuracy with decreasing training error without the presence of overfitting in any way. Therefore, these results demonstrate the viability of using a CNN-based model with multiple classes of lung disease.These results further demonstrate that when a CNN model is enhanced using transfer learning, the ability to produce an accurate and consistent classification of lung disease will improve significantly.

VII. CONCLUSION AND FUTURE SCOPE

Digital systems have enormous potential to be the next advanced tool in lung disease diagnosis by analyzing 3D CT scans with volumetric data to evaluate the location, severity, and spread of the disease. The use of explainable AI (XAI) makes the process more transparent by identifying the areas that are most influencing the diagnostic decision. Deploying cloud and edge services facilitates usage in real-time, e.g., in remote healthcare scenarios. The designed hybrid deep learning model that merges CNN and BiLSTM is the right way to find out the presence of lung diseases, including COVID-19, pneumonia, tuberculosis, and lung cancer, through the analysis of X-ray and CT images. CNN helps to extract the spatial features, and BiLSTM relates the sequential dependencies, thus outperforming traditional machine learning models in terms of accuracy and generalization.

REFERENCES

[1] Enshaei, N. (2025). Deep Learning-Based Medical Image Analysis for Enhanced Diagnosis and Severity Assessment of Viral Pneumonia (Doctoral dissertation, Concordia University).

[2] Rahman, N. N., Rahman, R., Jahan, N., & Adnan, M. A. (2024). Unveiling Diagnostic Precision: Evaluating Machine Learning and Deep Learning Approaches for Pneumonia Recognition of COVID-19 Patients Using Chest X-Rays. In Data-Driven Clinical Decision-Making Using Deep Learning in Imaging (pp. 61-81). Singapore: Springer Nature Singapore.

[3] Luz EJN ., et al., "Towards an Effective and Efficient Deep Learning Model for COVID-19 Patterns Detection in X-ray Images." *Research on Biomedical Engineering.* 2022; 38(1): 149–162. DOI: 10.1007/s42600-021-00112-7.

[4] Rahman T., et al. "Transfer Learning with Deep Convolutional Neural Network (CNN) for Pneumonia Detection Using Chest X-Ray." *Applied Sciences.* 2021; 11(7): 3233. DOI: 10.3390/app11073233.

[5] Ozturk T., et al, "Automated Detection of COVID-19 Cases Using Deep Neural Networks with X-ray Images." *Computers in Biology and Medicine.* 2020; 121: 103792. DOI: 10.1016/j.compbiomed.2020.103792.

[6] Akbar, W., Soomro, A., Hussain, A., Hussain, T., Ali, F., Haq, M. I. U., ... & Alsagri, R. (2024). Pneumonia detection: a comprehensive study of diverse neural network architectures using chest X-rays. International Journal of Applied Mathematics and Computer Science, 34(4), 679-699.

[7] P. Rajpurkar,, et al "CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning," arXiv preprint arXiv:1711.05225, Nov. 2017.

[8] X. Wang, et al "ChestX-ray8: Hospital-scale Chest X-ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases," arXiv preprint arXiv:1705.02315, May 2017

[9] J. Irvin, et al., "CheXpert: A Large Chest Radiograph Dataset with Uncertainty Labels and Expert Comparison," arXiv preprint arXiv:1901.07031, 2019.

[10] M. E. H. Chowdhury, et al "Can AI help in screening Viral and COVID-19 pneumonia?," IEEE Access, vol. 8, pp. 132665–132676, Jul. 2020, doi:10.1109/ACCESS.2020.3010287.

[11] M. F. Hashmi, et al "Efficient Pneumonia Detection in Chest X-ray Images Using Deep Transfer Learning," Diagnostics, vol. 10, no. 6, p. 417, Jun. 2020.

[12] R. Kundu, et al , "Pneumonia detection in chest X-ray images using an ensemble of CNN models," PLOS ONE, 2021

[13] E. Tiu, J. Carin, et al., "Expert-level detection of pathologies from unannotated chest X-rays via self-supervised learning," Nature Biomedical Engineering, 2022.

[14] S. B. Desai and colleagues, et al, "Deep learning and its role in COVID-19 medical imaging," Medical Image Analysis / review, 2020.

[15] M. R. Hasan, et al, "Recent advancement of deep learning techniques for pneumonia detection from chest X-rays: A survey (2022–2024)," Journal / review, 2024.